General Mathematics — Exercise Sheet 40

Line integrals, potentials and Green

Variant: Statements

Exercise 1 (Two paths and their difference). For $\boldsymbol{F} = (y^2, x)$, compute work from $(0,0)$ to $(1,1)$ along $\gamma_1(t) = (t, t^2)$ and $\gamma_2(t) = (t,t)$, $0 \le t \le 1$. Orient the closed path obtained from γ_1 and the reverse of γ_2, and verify the difference by Green's theorem on its enclosed region. Use Lecture 40 and the proper polynomial integrals of Lecture 26.

Exercise 2 (Constructing a nonpolynomial potential). For

$$\boldsymbol{F} = (e^x \cos y + 2xz,\ -e^x \sin y,\ x^2 + 3z^2)$$

on $\mathbb{R}^3$, verify the cross-partial hypotheses, construct a potential by successive component integration, and find work from $(0,0,0)$ to $(\ln 2, \pi, 1)$ along any piecewise C^1 path. Use Lecture 40's star-shaped theorem and Lecture 8's derivative rules.

Exercise 3 (A potential on a non-star-shaped domain). On $U = \mathbb{R}^2 \setminus \{0\}$ consider $\boldsymbol{F} = (2x/(x^2+y^2), 2y/(x^2+y^2))$. Find a single potential on all of U, and compute work from $(1,0)$ to $(0,2)$ along any piecewise C^1 path in U. Explain why this does not contradict the lecture's punctured-plane guard. Use Lecture 8's logarithmic derivative and Lecture 40's endpoint formula.

Exercise 4 (An exact differential equation and a local graph). An equation $M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$ is called exact if $M = \partial_x \Phi$, $N = \partial_y \Phi$ for one scalar potential Φ. For $M = 2xy + e^x$ and $N = x^2 + 2y$, find such a potential and show that the level through $(0,1)$ determines a unique local C^2 solution graph $y = y(x)$ of $y' = -M/N$. Compute $y'(0)$, $y''(0)$ and its quadratic Taylor approximation. Use the cited implicit theorem [C] in Lecture 37 and the endpoint/chain argument of Lecture 40; retain the existence dependence on [C].

Exercise 5 (Green on a curved cap). Let $D = \{0 \le x \le 1,\ 0 \le y \le x(1-x)\}$ and $\boldsymbol{F} = (-y^2, x^2 y)$. Compute positive boundary circulation by Green and verify it directly on the two boundary arcs. Check that D is simple in both directions. Use Lectures 26 and 40.

Exercise 6 (An annulus with two boundary components). Let $0 < a < b$, $D = \{a^2 \le x^2 + y^2 \le b^2\}$ and

$$\boldsymbol{F} = \left(-\frac{y}{x^2+y^2} - \frac{y}{2},\ \frac{x}{x^2+y^2} + \frac{x}{2} \right).$$

Compute the positively oriented integral on each boundary component and their sum. Compare with the finite-union/holes extension of Green [S], and explain why applying Green to the whole outer disc is invalid. Use Lecture 40's circle computation and orientation convention.